

0.1 Topology of R : Open/Closed Sets, Compact Sets, and the Heine-Borel Theorem

Having constructed the real numbers as a complete ordered field, we now formalize its topological structure. The standard topology on R is induced by the Euclidean metric $d(x, y) = |x - y|$.

Let $x \in R$ and $\epsilon > 0$. The ϵ -neighborhood of x is the open interval $V_\epsilon(x) = (x - \epsilon, x + \epsilon)$. A set $U \subset R$ is **open** if for every $x \in U$, there exists an $\epsilon > 0$ such that $V_\epsilon(x) \subset U$. A set $F \subset R$ is **closed** if its complement $F^c = R \setminus F$ is open.

A point $x \in R$ is a **limit point** of a set $A \subset R$ if every neighborhood $V_\epsilon(x)$ intersects A at a point distinct from x . That is, $(V_\epsilon(x) \setminus \{x\}) \cap A \neq \emptyset$ for all $\epsilon > 0$.

A set $F \subset R$ is closed if and only if it contains all its limit points.

Suppose F is closed, and let x be a limit point of F . We proceed by contradiction. If $x \notin F$, then $x \in F^c$. Because F^c is open by definition, there exists $\epsilon > 0$ such that $V_\epsilon(x) \subset F^c$. This implies $V_\epsilon(x) \cap F = \emptyset$, which contradicts the premise that x is a limit point of F . Therefore, $x \in F$.

Conversely, suppose F contains all its limit points. We must show that F^c is open. Let $y \in F^c$. Since $y \notin F$, y is not a limit point of F . By definition, there exists an $\epsilon > 0$ such that $V_\epsilon(y)$ contains no points of F other than possibly y itself. But $y \notin F$, so $V_\epsilon(y) \cap F = \emptyset$. This means $V_\epsilon(y) \subset F^c$. Since this holds for all $y \in F^c$, F^c is open, and thus F is closed.

An **open cover** for a set $E \subset R$ is a collection of open sets $\{G_\alpha\}_{\alpha \in A}$ such that $E \subset \bigcup_{\alpha \in A} G_\alpha$. A set $K \subset R$ is **compact** if every open cover of K contains a finite subcover.

[Heine-Borel Theorem] A subset $K \subset R$ is compact if and only if it is closed and bounded.

() Assume K is compact. We first show K is bounded. Consider the open cover consisting of intervals $I_n = (-n, n)$ for all $n \in N$. Since $K \subset \bigcup_{n=1}^{\infty} I_n$, compactness guarantees a finite subcover $\{I_{n_1}, I_{n_2}, \dots, I_{n_k}\}$. Let $N = \max\{n_1, \dots, n_k\}$. Then $K \subset (-N, N)$, meaning K is bounded.

Next, we show K is closed by proving K^c is open. Let $p \in K^c$. For each $q \in K$, choose disjoint open neighborhoods U_q of p and V_q of q ; explicitly, one may take $\epsilon = \frac{|p-q|}{2}$ and let $U_q = V_\epsilon(p)$ and $V_q = V_\epsilon(q)$. The collection $\{V_q\}_{q \in K}$ is an open cover of K . By compactness, there exists a finite subcover $\{V_{q_1}, \dots, V_{q_m}\}$. Let $W = \bigcap_{i=1}^m U_{q_i}$. As a finite intersection of open sets, W is open. Furthermore, $p \in W$, and $W \cap V_{q_i} = \emptyset$ for all $i = 1, \dots, m$. Because $\{V_{q_i}\}$ covers K , it follows that $W \cap K = \emptyset$. Thus, $W \subset K^c$, proving K^c is open and K is closed.

() Assume K is closed and bounded. Because K is bounded, $K \subset [a, b]$ for some real numbers $a < b$. We first prove $[a, b]$ is compact. Let $\{G_\alpha\}_{\alpha \in A}$ be an open cover of $[a, b]$. Define the set S as:

$$S = \{x \in [a, b] \mid \text{the interval } [a, x] \text{ is covered by a finite subcollection of } \{G_\alpha\}\}$$

Since $a \in [a, b]$, there is some G_{α_0} containing a , meaning $[a, a]$ is finitely covered. Thus $a \in S$, and $S \neq \emptyset$. Furthermore, S is bounded above by b . By the Completeness Axiom established in Section 1.1, S possesses a supremum $c = \sup S$, where $a \leq c \leq b$.

Since $c \in [a, b]$, $c \in G_\beta$ for some $\beta \in A$. Because G_β is open, there exists $\epsilon > 0$ such that $(c - \epsilon, c + \epsilon) \subset G_\beta$. By the definition of the supremum, there exists $x \in S$ such that $c - \epsilon < x \leq c$. Because $x \in S$, $[a, x]$ is covered by a finite subcollection $\mathcal{F} \subset \{G_\alpha\}$. The augmented collection $\mathcal{F} \cup \{G_\beta\}$ is finite and covers $[a, c + \frac{\epsilon}{2}]$. If $c < b$, this implies $c + \frac{\epsilon}{2} \in S$, contradicting $c = \sup S$. Hence, $c = b$, and by the same logic, $b \in S$. Therefore, $[a, b]$ is finitely covered, making it compact.

Finally, let $\{U_\lambda\}_{\lambda \in \Lambda}$ be an open cover of K . Since K is closed, K^c is open. The collection $\{U_\lambda\}_{\lambda \in \Lambda} \cup \{K^c\}$ forms an open cover of the compact set $[a, b]$. This extended cover admits a finite subcover Φ . Removing K^c from Φ leaves a finite subcollection of $\{U_\lambda\}$ that still fully covers K . Thus, K is compact.